

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration: 1 Hour
Total Marks: 50

Annexure A

Scenario Based Assignment

Code of Conduct:

(192524118)

I R. Akshaya (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:







RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

$20 \times 20 + 23 + 18 + 13$
 $= 94$


1. Scenario

A search-and-rescue drone is deployed inside a building that has partially collapsed after an earthquake. It must: navigate safely to locate survivors, avoid hazards (unstable debris, gas leaks), operate with limited sensing range, and make decisions without a complete structural map. The drone updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i. Explain why this problem requires an online search agent instead of offline search.
- ii. Analyze the challenges of partial observability and dynamic environments.
- iii. Describe how the drone builds and updates its internal model.
- iv. Compare online search with uninformed and informed search strategies.
- v. Justify the most suitable approach considering uncertainty, adaptability, and safety.

2. Scenario

An event management company is designing an AI system to automatically generate seating arrangements for a large wedding banquet. The system must satisfy multiple constraints: no two guests with known conflicts should be seated at the same table, a limited number of seats are available per table, certain family groups must be seated together, guests with dietary/accessibility needs require specific table locations, and special accommodations for VIP guests near the head table. The complexity increases as the number of guests and tables grows.

Tasks:

Formulate the problem as a Constraint Satisfaction Problem (CSP).

- i. Clearly define variables, domains, and constraints with explanation.
- ii. Explain how backtracking search works in this scenario.
- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency.
- iv. Analyze real-world challenges and justify the best strategy for scalability.

3. Scenario

A fintech startup is developing an AI agent to compete in a simulated auction-bidding game against other automated bidders. The AI must compete against adversarial bidding agents, make decisions within strict time limits per bidding round, evaluate complex market states with multiple possible bids, and the decision tree of possible bid sequences is extremely large, making computation expensive.

Tasks:

- i. Explain how the Minimax algorithm models this problem.
- ii. Analyze the computational challenges of large game/decision trees.
- iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning.
- iv. Design an evaluation function and justify the factors considered.
- v. Recommend a strategy for balancing optimality and real-time performance.

1. SEARCH- AND- RESCUE DRONE

(i) Why online search instead of offline search?

⇒ An online search instead agent is needed because the drone does not know the complete building structure in advance. It discovers obstacles, survivors, hazards while exploring.

⇒ Therefore, it must sense, act, learn and re-plan continuously.

(ii) Challenges of partial observability and dynamic Environments :

⇒ Limited sensor range and incomplete information.

⇒ Survivors and obstacles may be hidden.

⇒ Debris or structures may change.

⇒ Gas leaks and unsafe areas may appear.

⇒ Sensor readings may be uncertain or noisy.

(iii) How the drone updates its internal model.

→ The drone collects information using sensors and builds an internal map containing obstacles, safe path, hazards, explored areas, and survivor locations.

→ As new information is received, the map is updated and the route is re-planned.

(iv) Online VS uninformed and informed search.

⇒ Uninformed search: Does not use additional knowledge; Eg: BFS and DFS.

⇒ Informed search: Uses heuristic,
Eg: A^*

⇒ Online search: Works with incomplete information and updates decision while exploring.

⇒ Online search is more suitable because the environment is unknown and changing.

(v) Most suitable approach:

→ An online, heuristic-based search with continuous sensing and re-planning is most suitable.

→ Safety should be given higher priority than the shortest path, and hazardous areas should have high costs.

2. WEDDING SEATING ARRANGEMENT-CSP.

(i) Variables, domains and constraints:

Variables: Each guest represents a variable.

Domains: Possible tables for each guest.

Example: $G_1 = \{T_1, T_2, T_3\}$

Constraints:

- * Conflicting guests cannot sit at the same table.
- * Table capacity cannot be exceeded.
- * Family groups should sit together.
- * VIPs should be near the table.
- * Accessibility and dietary requirements must be satisfied.

(ii) Backtracking Search:

⇒ Backtracking assigns guests to tables one by one.

⇒ If an assignment violates a constraint, the algorithm goes back and tries another table.

Eg: $G_1 \rightarrow T_1 \rightarrow G_2 \rightarrow \text{conflict} \rightarrow \text{backtrack} \rightarrow G_2 \rightarrow T_2$,

(iii) Constraint Propagation:

Constraint propagation removes invalid choices early.

Eg: If G_1 and G_2 have a conflict and G_1 is assigned to T_1 , T_2 is removed from G_2 's domain. This reduces unnecessary searching and improves efficiency.

(iv) Real-world challenges & scalable:

⇒ The no. of possible arrangements becomes very large with more guests and tables.

⇒ Other challenges include conflicts, VIP requirements & last-minute changes.

A good strategy is

Backtracking + Constraint Propagation + MRV + LCV

It reduces search space and improves scalability.

3. AI AUCTION - BIDDING GAME.

(i) Minimax Algorithm:-

The auction can be represented as a game tree where each node represents a market state and each branch represents a possible bid.

The AI acts as MAX, trying to maximize profit, while opponent acts as MIN, trying to reduce the AI's benefit.

(ii) Computational Challenges:

- * Large game trees create high computational costs.

- * If branching factor is b and depth is d , the no. of nodes can be approximately:
 $O(b^d)$

- * The AI also has strict limits & must handle many possible bids & opponent actions

(iii) Alpha-Beta pruning:

⇒ It removes the branches that cannot affect final Minimax decision.

Alpha: Best value found for Max.

Beta: Best value found for Min.

Where $\alpha \geq \beta$, further exploration stopped.

∴ It gives the same Minimax result.

(iv) Evaluation function:

Evaluation = Profit + Future value - Bid Cost - Risk + Resource Advantage.

Important factors are current profit, future opportunities, bidding cost, risk, remaining budget.

(v) Best strategy :-

Use :

Depth - Limited Minimax + Alpha-Beta Pruning + Heuristic Evaluation + Iterative Deepening.

This provides good decision meeting real time requirements. It balances optimality, speed, efficiency.